Answer Key — AP Statistics Class 11: Random Variables & Expected Value

SOMATH — Teacher's edition. 60 solutions across 12 concepts.

1. What Is a Random Variable?

Q1. Yes. X is a numerical value determined by a random process. Possible values: **1, 2, 3, 4**.

Q2. **Possible values: 0 and 1.** This is called an indicator random variable.

Q3. **0, 1, 2, 3, 4** — any whole number from 0 up to the number of flips.

Q4. **False.** A random variable is any numerical value determined by chance. Height, time, and temperature are all random variables that can take non-whole values.

Q5. **Not quite.** A random variable must be *numerical*. To turn color into a random variable, we'd assign numbers (for example 1 = red, 0 = blue).

2. Discrete vs. Continuous Random Variables

Q1. **Discrete** — you count cars (0, 1, 2, …).

Q2. **Continuous** — time can take any value in an interval (10.24 s, 10.241 s, …).

Q3. **Discrete** — a countable whole number.

Q4. **Continuous** — weight is measured, not counted.

Q5. **Discrete.** The possible values can be listed — they don't form a continuum.

3. Probability Distribution of a Discrete RV

Q1. Every value 1–6 has probability $1/6$. Distribution: **P(X = k) = 1/6 for k = 1, 2, 3, 4, 5, 6**.

Q2. Outcomes: HH, HT, TH, TT. **P(X = 0) = 1/4, P(X = 1) = 1/2, P(X = 2) = 1/4**.

Q3. $P(X \geq 2) = P(X = 2) + P(X = 3) = 0.5 + 0.3 = \mathbf{0.8}$.

Q4. $P(X < 3) = P(X = 1) + P(X = 2) = 0.2 + 0.5 = \mathbf{0.7}$.

Q5. **(1)** Identify the random variable; **(2)** list all possible values; **(3)** find the probability of each value; **(4)** check that the probabilities add to 1.

4. Rules for a Valid Probability Distribution

Q1. 0.3 + 0.5 + 0.2 = 1.0 and each is in [0, 1]. **Valid.**

Q2. Sum = 1.1 ≠ 1. **Not valid** — violates Rule 2.

Q3. Sum = 1.00 and each is in [0, 1]. **Valid** (uniform on 4 values).

Q4. **Not valid** — P(X = 3) = −0.1 violates Rule 1, even though the sum is 1.0.

Q5. **(1)** Each probability satisfies $0 \le P(X = x) \le 1$. **(2)** The probabilities sum to 1: $\Sigma\, P(X = x) = 1$.

5. Finding Missing Probabilities

Q1. 0.1 + 0.4 + P + 0.2 = 1 ⇒ P = 1 − 0.7 = **0.3**.

Q2. 0.15 + P + 0.55 = 1 ⇒ P = 1 − 0.70 = **0.30**.

Q3. Sum of known = 0.60. P = 1 − 0.60 = **0.40**.

Q4. The sum of the given probabilities is already greater than 1, so the table is inconsistent. Check the given values — at least one is wrong.

Q5. Let $P(Z = 4) = p$. Then $P(Z = 3) = 2p$. Sum: 0.2 + 0.2 + 2p + p = 1 ⇒ 3p = 0.6 ⇒ p = 0.2. So **P(Z = 3) = 0.4, P(Z = 4) = 0.2**.

6. Expected Value — The Mean of a Random Variable

Q1. $E(X) = 0(0.2) + 1(0.5) + 2(0.3) = 0 + 0.5 + 0.6 = \mathbf{1.1}$.

Q2. $E(X) = 1(0.5) + 2(0.3) + 5(0.2) = 0.5 + 0.6 + 1.0 = \mathbf{2.1}$.

Q3. Net values: +\$8 (win) or −\$2 (lose). $E = 8(0.1) + (-2)(0.9) = 0.8 - 1.8 = \mathbf{-\$1.00}$. Expect to lose \$1 per play.

Q4. $E(X) = 100(0.5) + 200(0.3) + 300(0.2) = 50 + 60 + 60 = \mathbf{170}$.

Q5. $E(\text{payout}) = 500(0.02) + 0(0.98) = \mathbf{\$10}$. The break-even premium is \$10 (before insurer expenses/profit).

7. Interpreting Expected Value

Q1. **No** — you can only observe 0, 1, 2, 3, or 4 heads. But over many sets of 4 flips, the average number of heads is 2.7.

Q2. Expected value is the **weighted average** of all outcomes (each weighted by probability). Most likely = the value with the highest probability. They can be different (and often are).

Q3. **Not necessarily.** Any single day's profit could be higher or lower. Over many days, the average will be close to \$450.

Q4. If you repeat the "flip 3 coins" experiment many times, the **average** number of heads per set of 3 flips will be about 1.5.

Q5. **False.** E(X) can be any real number in the range of X, including non-attainable values like 3.5 on a die or 1.5 heads in 3 flips.

8. Expected Value in Games and Decisions

Q1. $E(X) = 5(0.3) + (-2)(0.7) = 1.5 - 1.4 = \mathbf{\$0.10}$. Positive — slightly favorable in the long run.

Q2. $E(X) = 10(0.15) + (-3)(0.85) = 1.5 - 2.55 = \mathbf{-\$1.05}$. Negative — unfavorable, don't play.

Q3. $E(\text{no warranty cost}) = 400(0.03) = \$12$. With warranty guaranteed cost is \$20. \$12 < \$20, so by expected value **skip the warranty**.

Q4. $E(\text{no warranty cost}) = 400(0.10) = \$40$. \$40 > \$20, so **buy the warranty**.

Q5. $E(B) = 20(0.6) + 45(0.4) = 12 + 18 = \mathbf{30\ min}$. Same expected time as A, but B has more variability. On expected value alone, they're tied.

9. Variance of a Random Variable

Q1. $\mu = 1.1$. Deviations squared: $(0-1.1)^2 = 1.21$, $(1-1.1)^2 = 0.01$, $(2-1.1)^2 = 0.81$. Var = 1.21(0.2) + 0.01(0.5) + 0.81(0.3) = 0.242 + 0.005 + 0.243 = **0.49**.

Q2. **0.** There is no variation from the mean.

Q3. Both means are 0. Var(A) = 100, Var(B) = 4. **Game A** is more spread out.

Q4. **No.** It is a sum of squared deviations weighted by non-negative probabilities, so it is always ≥ 0.

Q5. $\mu = 2$. Deviations squared: 1 and 1. Var = 1(0.5) + 1(0.5) = **1**.

10. Standard Deviation of a Random Variable

Q1. $\sigma_X = \sqrt{16} = \mathbf{4}$.

Q2. $\sigma_X = \sqrt{0.49} = \mathbf{0.7}$.

Q3. $\text{Var} = 2.5^2 = \mathbf{6.25}$.

Q4. $\sigma_X = \sqrt{0.75} \approx \mathbf{0.87}$ heads.

Q5. Because standard deviation is in the **same units** as the variable itself (dollars, minutes, cm), so it's directly interpretable. Variance is in squared units.

11. Transforming Random Variables

Q1. $\mu_Y = 7 + 10 = \mathbf{17}$. $\sigma_Y = \mathbf{3}$ (unchanged by shift).

Q2. $\mu_Y = 4(10) = \mathbf{40}$. $\sigma_Y = |4|(3) = \mathbf{12}$.

Q3. $\mu_Y = 3 + 2(20) = \mathbf{43}$. $\sigma_Y = |2|(5) = \mathbf{10}$.

Q4. $\mu_Y = 100 + (-1)(50) = \mathbf{50}$. $\sigma_Y = |-1|(8) = \mathbf{8}$.

Q5. Standard deviation is always **nonnegative**. The rule uses $|b|$, not b : $\sigma_Y = |-2|\cdot\sigma_X = 2\cdot\sigma_X$, which is positive.

12. Combining Random Variables

Q1. $\mu_{X+Y} = 14$. $\mu_{X-Y} = \mathbf{6}$. Means always add or subtract.

Q2. $\sigma_{X+Y} = \sqrt{(3^2 + 4^2)} = \sqrt{25} = \mathbf{5}$.

Q3. Variances still ADD for a difference: $\sigma_{X-Y} = \sqrt{(9 + 16)} = \mathbf{5}$ (same as sum).

Q4. Each die: $\mu = 3.5, \sigma \approx 1.71$. Sum: $\mu = 7$, $\sigma = \sqrt{(1.71^2 + 1.71^2)} \approx \mathbf{2.42}$.

Q5. **No.** Variances add, standard deviations do NOT. Use $\sigma_{X+Y} = \sqrt{(\sigma_X^2 + \sigma_Y^2)}$.